

Question:

John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for \$15 each. He ends up buying 8 video games for \$20 each and has \$5 left. How many lego sets does he still have?

Reasoning + Answer:

Let's go through the problem step by step .

John plans to sell all his lego sets and use the money to buy video games . He has 13 lego sets and sells them for \$ 15 each .

So the total money he gets from selling the legos is :

\$\$
 $13 \times 15 = 195 \text{ \texttt{\text{ dollars }}}$
\$\$

He then buys 8 video games for \$ 20 each :

\$\$
 $8 \times 20 = 160 \text{ \texttt{\text{ dollars }}}$
\$\$

He has \$ 5 left after the purchase . That means :

\$\$
 $\text{Money spent} = 160$
\$\$
\$\$
 $\text{Money left} = 5$
\$\$
\$\$
 $\text{Total money available} = 160 + 5 = 165$
\$\$

Wait â thatâs inconsistent . He got \$ 195 from selling the legos , and we now see that if he spent \$ 160 and had \$ 5 left , he would have 165 total money . But \$ 195 - \$ 165 = \$ 30 left .

So something 's wrong â the problem says he ends up buying 8 video games for \$ 20 each and has \$ 5 left . So :

Total money available = money from selling legos = $13 \times 15 = \$195$

Money spent on games = $8 \times 20 = \$160$

Money left = \$ 5

So total money used = $160 + 5 = \$165$

But he made \$ 195 â He had \$ 195 , spent \$ 165 â \$ 30 left ? Contradiction .

Wait â so the statement " he ends up buying 8 video games for \$ 20 each and has \$ 5 left " implies that the total amount of money he gets from selling legos is exactly :

\$\$
 $\text{Money from legos} = 8 \times 20 + 5 = 160 + 5 = 165$
\$\$

So he earned \$ 165 from selling legos .

He had 13 lego sets , each for \$ 15 â total money from selling = $13 \times 15 = \$195$

So \$ 195 , not \$ 165 . So he couldn't have only sold 13 sets and have only \$ 165 . Unless...

Wait â The key point of the question is : â How many lego sets does he still have ?â

So maybe he didnât sell **all** of them ? The question says "John plans to sell all his toys..." but then the calculation leads to a contradiction unless he didn't sell all of them .

Letâs re-read the question :

"John plans to sell all his toys and use the money to buy video games . He has 13 lego sets and he sells them for \$ 15 each . He ends up buying 8 video games for \$ 20 each and has \$ 5 left . How many lego sets does he still have ?"

So it says : " he sells them " â the â them â refers to â his toys â , which are the lego sets .

He had 13 . He sells them for \$ 15 each â total money = $13 \times 15 = \$195$

Then he buys 8 games at \$ 20 â $8 \times 20 = \$160$

Money left : \$ 5 â so total expenses = $160 + 5 = 165$

But thatâs inconsistent : he had \$ 195 , spent \$ 165 â should have \$ 30 left .

But the problem says he has \$ 5 left .

So either the initial premise is false , or the interpretation is wrong .

Wait â maybe the question says " he sells them " meaning he sells *some